

JEE MAIN REVISION MASTER: MOTION IN A PLANE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 90 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Projectile Motion & Trajectory	74 PYQs	Projectiles
Very High	Relative Velocity in 2D (Rain-Man & River-Swimmer)	16 PYQs	Relative Motion

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Projectile Motion & Trajectory	$T = \frac{2u \sin \theta}{g}$, $H = \frac{u^2 \sin^2 \theta}{2g}$, $R = \frac{u^2 \sin 2\theta}{g}$, $y = x \tan \theta \left(1 - \frac{x}{R}\right)$
Relative Velocity in 2D (Rain-Man & River-Swimmer)	$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$, $t_{\text{min}} = \frac{d}{v_{\text{br}}}$, $\text{Drift} = (v_r - v_{\text{br}} \sin \theta) t$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Projectile Motion & Trajectory (Very High Yield — 74 PYQs)

Core Principle: 2D motion under constant gravity with independent horizontal and vertical components.

■ **Caution / JEE Pitfall:** Range is identical for complementary angles (θ and $90^\circ - \theta$), but flight time and maximum height differ.

■ **Speed Trick:** Equation of trajectory form $y = x \tan \theta (1 - x/R)$ allows fast calculation of range when trajectory coordinates are given.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R . If the time of flight of $t...$ " [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "A boy throws a ball into air at 45° from the horizontal to land it on a roof of a building of height H . If the ball attains maxi..." [Ans: D]
- [2026 Evening Shift] Q#2 (MCQ): "At $t=0$, a body of mass 100 g starts moving under the influence of a force $(5\hat{i} + 10\hat{j})\text{ N}$. After 2 s its position is $(2x\hat{i} + \dots)$ " [Ans: D]

3.2 Relative Velocity in 2D (Rain-Man & River-Swimmer) (Very High Yield — 16 PYQs)

Core Principle: Relative vector addition in planar motion, umbrella holding angle for rain, and river crossing optimization.

■ **Caution / JEE Pitfall:** To cross a river in shortest time, head perpendicular to the river flow ($\theta = 90^\circ$ relative to bank).

■ **Speed Trick:** To cross with zero drift (shortest path), swim upstream such that $\sin\theta = \frac{v_{\text{river}}}{v_{\text{swimmer}}}$ (requires $v_{\text{swimmer}} > v_{\text{river}}$).

Representative Source PYQs:

- **[2026 Evening Shift] Q#3 (MCQ):** "A river of width 200 m is flowing from west to east with a speed of 18 km/h. A boat, moving with speed of 36 km/h in still water, is made to..." **[Ans: C]**
- **[2025 Morning Shift] Q#1 (MCQ):** "A river is flowing from west to east direction with speed of 9 km h^{-1} . If a boat capable of moving at a maximum speed of 27 km h^{-1} in..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#3 (MCQ):** "The position vector of a moving body at any instant of time is given as $\vec{r}(t) = (5t^2\hat{i} - 5t\hat{j})\text{ m}$. The magnitude and direction o..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R . If the time of flight of these bodies are 5 s and 10 s, respectively, then the value of R is ____ m. (Take $g = 10\text{ m/s}^2$)

- (A) 250 **[Correct]**
- (B) 25
- (C) 500
- (D) 125

Correct Answer: A

Solution Analysis: For two projectiles launched with the same initial speed u to have the same horizontal range R , their angles of projection must be complementary. Let the first angle be θ . Then the second angle must be $(90^\circ - \theta)$. The time taken for the projectile to return to the same horizontal level. $T = (2u \sin\theta / g)$... (i) And the total horizontal distance covered is the range of projectile, R . $R = \frac{u^2 \sin(2\theta)}{g} = \frac{2u^2 \sin\theta \cos\theta}{g} \Rightarrow R = \frac{2u^2 \sin\theta \cos\theta}{g}$ (ii) For the first body ($T_1 = 5\text{ s}$): $T_1 = \frac{2u \sin\theta}{g}$

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Evening Shift (Q#3)

A river of width 200 m is flowing from west to east with a speed of 18 km/h. A boat, moving with speed of 36 km/h in still water, is made to travel one-round trip (bank to bank of the river). Minimum time taken by the boat for this journey and also the displacement along the river bank are _____ and _____ respectively.

- (A) 20 s and 100 m
- (B) 40 s and 100 m
- (C) 40 s and 200 m **[Correct]**
- (D) 40 s and 0 m

Correct Answer: C

Solution Analysis: The speed of river is $v_r = 18 \times (5 / 18) = 5 \text{ m/s}$ The speed of boat in still water is $v_b = 36 \times (5 / 18) = 10 \text{ m/s}$ The width of river is $w = 200 \text{ m}$ The time taken to cross a river is determined solely by the component of the boat's velocity perpendicular to the river banks. $t = \frac{w}{v_b \sin \theta}$ Where θ is the angle with the bank. Time is minimum when the boat heads straight across ($\theta = 90^\circ$), so $\sin \theta = 1$. $t_{\min 1 \rightarrow 2} = \frac{w}{v_b} = (200 / 10) = 20 \text{ s}$ Also, while returning to initial bank in minimum time, $t_{\min 2 \rightarrow 1} = \frac{w}{v_b}$